

JASPER WARDEN, AB, Canada

WMO: 714860

Lat: 52.9267N

Lon: 118.0297W

Elev: 1020

StdP: 89.66

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -31.1	(c) -26.8	(d) -35.2	(e) 0.2	(f) -30.6	(g) -31.1	(h) 0.2	(i) -26.0	(j) 6.3	(k) -15.0	(l) 5.4	(m) -13.4	(n) 1.3	(o) 190	(p) 0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.4	(c) 29.4	(d) 14.5	(e) 27.0	(f) 13.8	(g) 24.9	(h) 13.1	(i) 15.6	(j) 25.7	(k) 14.7	(l) 24.3	(m) 13.9	(n) 22.6	(o) 2.4	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	10.3	16.0	11.5	9.6	15.5	10.5	8.9	15.0	46.9	25.8	44.3	24.4	42.0	22.8	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	4.9	4.2	-35.8	32.5	3.8	1.8	-38.6	33.8	-40.8	34.9	-43.0	35.9	-45.8	37.2		
			-36.0	17.2	3.7	1.1	-38.7	18.0	-40.9	18.7	-43.0	19.3	-45.7	20.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-8.6	-6.5	-1.9	3.4	8.4	12.4	15.3	14.4	10.0	3.8	-3.2	-9.0	(6)
(7)		DBStd	10.10	8.40	7.29	6.19	3.77	3.66	2.87	3.07	3.06	3.80	4.20	6.29	7.56	(7)
(8)		HDD10.0	2925	577	463	368	199	75	9	2	3	47	195	397	590	(8)
(9)		HDD18.3	5521	836	696	626	448	309	181	103	126	252	450	647	848	(9)
(10)		CDD10.0	458	0	0	0	1	24	79	166	139	46	3	0	0	(10)
(11)		CDD18.3	13	0	0	0	0	0	1	8	3	1	0	0	0	(11)
(12)		CDH23.3	806	0	0	0	1	24	87	357	268	67	1	0	0	(12)
(13)		CDH26.7	227	0	0	0	0	1	19	119	75	13	0	0	0	(13)
(14)	Wind	WSAvg	1.6	1.6	1.6	1.8	1.8	1.7	1.6	1.5	1.5	1.5	1.5	1.4	(14)	
(15)	Precipitation	PrecAvg	415	33	19	17	23	32	53	56	53	40	32	29	28	(15)
(16)		PrecMax	539	98	62	47	49	97	109	138	130	110	70	62	87	(16)
(17)		PrecMin	327	2	1	2	2	9	17	8	17	15	8	5	6	(17)
(18)		PrecStd	60	24	15	11	12	18	25	33	29	19	16	15	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	8.8	14.4	21.1	26.0	29.2	32.7	31.6	28.6	20.5	11.6	6.0	(19)
(20)			MCWB	2.6	2.8	5.6	9.4	11.6	13.8	15.3	15.5	13.4	10.0	6.2	2.3	(20)
(21)		2%	DB	4.8	6.5	10.6	17.1	23.3	25.7	29.9	28.9	25.3	16.2	7.9	3.7	(21)
(22)			MCWB	1.5	1.7	3.8	7.6	10.7	12.9	14.7	14.6	12.9	8.3	4.3	1.0	(22)
(23)		5%	DB	3.2	4.6	8.3	14.1	20.5	23.3	27.5	26.5	22.0	13.1	5.8	2.0	(23)
(24)			MCWB	0.6	0.4	2.7	6.1	9.9	12.0	14.1	13.9	11.9	7.2	3.0	-0.1	(24)
(25)		10%	DB	1.5	2.7	6.5	11.6	18.0	20.8	24.8	23.9	18.3	10.6	4.0	0.5	(25)
(26)			MCWB	-0.5	-0.5	1.5	4.6	8.9	11.1	13.5	13.3	10.4	5.9	1.7	-1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	3.7	6.2	9.7	12.7	15.0	16.7	16.8	14.5	11.0	6.8	2.8	(27)
(28)			MCD B	5.6	7.1	12.8	19.7	22.6	25.5	26.8	26.2	24.7	18.4	10.3	4.7	(28)
(29)		2%	WB	1.9	2.1	4.4	8.0	11.5	13.8	15.7	15.6	13.3	9.4	4.9	1.3	(29)
(30)			MCD B	4.1	5.6	9.2	15.7	21.0	22.6	25.9	25.6	23.3	14.8	7.5	3.2	(30)
(31)		5%	WB	0.8	0.9	3.1	6.5	10.5	12.8	14.8	14.7	12.2	7.7	3.4	0.1	(31)
(32)			MCD B	2.9	3.9	7.5	13.0	19.3	20.9	24.8	24.2	20.7	12.2	5.3	1.9	(32)
(33)		10%	WB	-0.4	-0.3	2.0	5.1	9.4	11.8	14.0	13.8	11.0	6.4	1.9	-1.2	(33)
(34)			MCD B	1.4	2.5	5.9	10.9	17.0	19.0	22.9	22.3	17.8	10.1	3.7	0.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.7	12.5	12.0	13.6	14.7	14.0	15.4	16.1	14.4	11.6	8.9	10.0	(35)
(36)			MCD BR	9.6	12.9	14.3	19.1	21.3	20.3	22.6	22.5	16.6	10.0	8.7	(36)	
(37)			MCWBR	6.6	8.4	8.3	10.4	10.3	8.8	8.9	9.5	10.7	9.3	6.6	6.4	(37)
(38)		5% WB	MCD BR	8.9	12.0	13.2	17.4	19.3	16.9	18.8	19.6	20.0	14.9	9.2	8.4	(38)
(39)			MCWBR	6.3	8.0	8.0	9.7	9.4	7.6	7.8	8.8	9.7	8.8	6.4	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.264	0.298	0.346	0.365	0.388	0.367	0.368	0.336	0.299	0.263	0.252	(40)	
(41)		taud	2.335	2.381	2.289	2.180	2.272	2.277	2.386	2.373	2.454	2.512	2.406	2.262	(41)	
(42)		Ebn at Noon	749	865	892	883	883	863	876	854	844	814	738	671	(42)	
(43)		Edh at Noon	57	75	101	129	125	126	111	106	87	66	53	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.81	1.67	2.71	4.16	4.96	5.04	5.34	4.59	3.17	1.77	0.85	0.63	(44)	
(45)		RadStd	0.08	0.17	0.26	0.39	0.49	0.44	0.46	0.28	0.37	0.19	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page